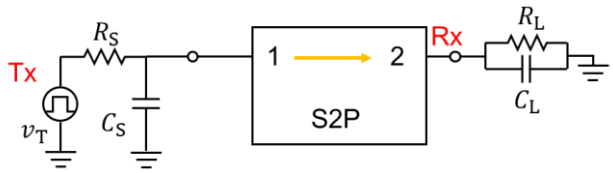
Homework: Jitter fitting and BER extrapolation

In the last homework, we have simulated the jitter distribution of the following channel (40 Gbps NRZ).



$V_{pp} = 1$ V	$T_b = 25$ ps	$R_S = 20\ \Omega$	$R_L = 85\ \Omega$
$T_d = 0$ ps	$T_r = 5$ ps	$C_S = 0.3$ pF	$C_L = 0.3$ pF

We have simulated four cases, each obtaining its Tx jitter PDF and Rx jitter PDF. Specifically,

- Case 1: Ideal Tx, no jitter. Namely, Rx only has data dependent jitter (DDJ).
- Case 2: Tx has sinusoidal jitter (SJ), with amplitude = 3 ps and frequency = 1 MHz.
- Case 3: Tx has random jitter (RJ), with zero mean and standard deviation σ = 1 ps.
- Case 4: Tx has both SJ and RJ, with the same parameters as Cases 2 and 3.

In this homework, we will fit the jitter distribution with the two-Gaussian mixture model, and use it to extrapolate the bathtub curve to obtain the eye width at low BER and the lowest possible BER.

Steps:

1. Follow the last homework to obtain the sampled jitter PMF (not PDF yet) of all cases.
2. Compute and plot the sampled jitter CDF (*sampled* means non-extrapolated, solely based on data).
3. Compute and plot the sampled bathtub (BER) curve of all cases (8 curves).

   Below, specifically focus on Case 3 and Case 4 Rx.
4. Scale the jitter PMF to make it a sampled PDF.
5. Pick a set of five parameters (μ_1, μ_2, σ_1, σ_2, τ_1), and $\tau_2 = 1 - \tau_1$. Compute the 2-Gaussian PDF:

$$\mathrm{PDF2G}(t) = \frac{\tau_1}{\sigma_1\sqrt{2\pi}} e^{\frac{-(t-\mu_1)^2}{2\sigma_1^2}} + \frac{\tau_2}{\sigma_2\sqrt{2\pi}} e^{\frac{-(t-\mu_2)^2}{2\sigma_2^2}}$$

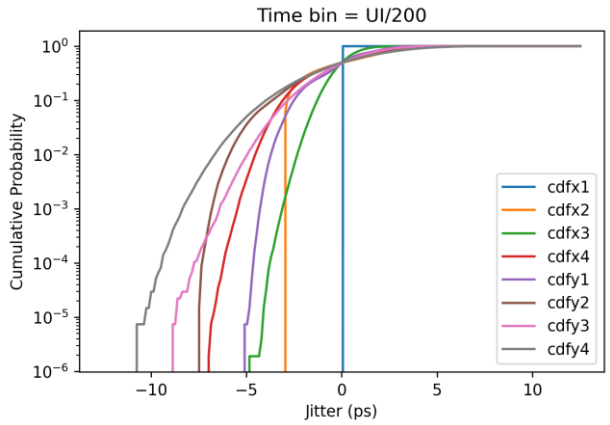
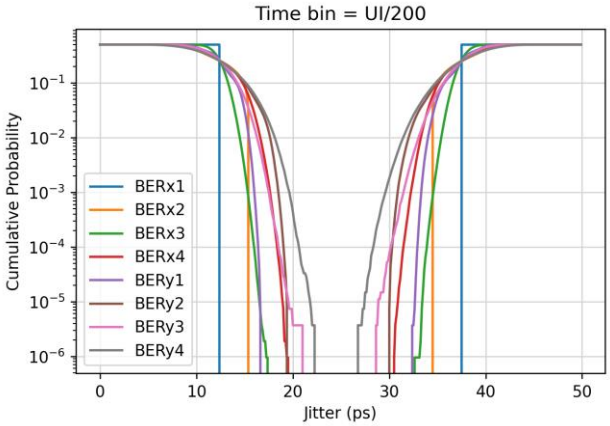
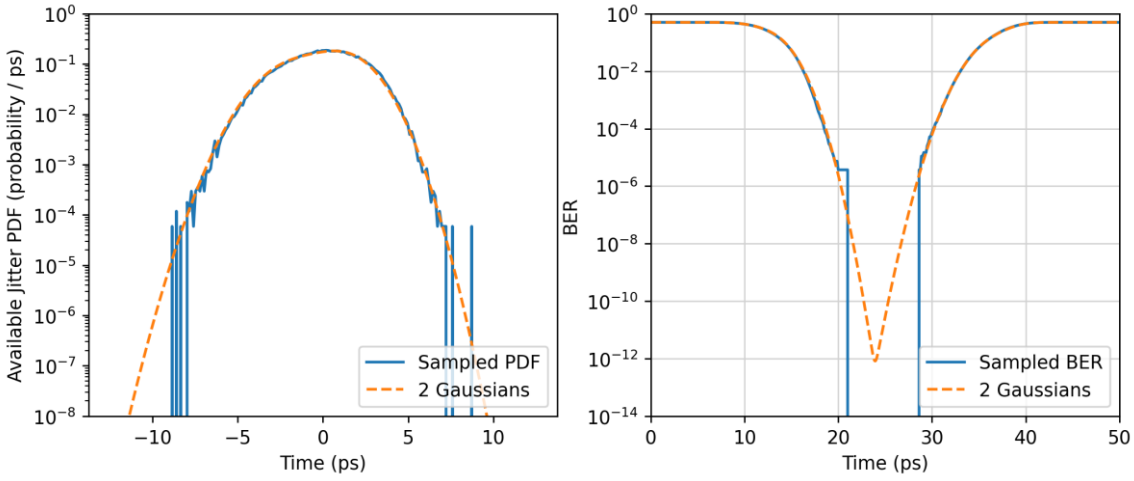
   and its corresponding CDF2G(t). (Hint: in the class note, we have derived the CDF of Gaussian distribution. The CDF for 2-Gaussian can be easily obtained based on it.) Then, based on the CDF2G, compute the BER curve.
6. Plot the sampled PDF against the 2-Gaussian PDF, and plot the sampled BER (bathtub) against the 2-Gaussian BER. Tune the five parameters such that the 2-Gaussian model fits the two tails of the sampled PDF/BER well. Note that you *only* have to fit the tail parts; the center part of jitter distribution is mainly caused by DJ, does not follow Gaussian distribution in any sense, and thus needs not and shall not be fitted by Gaussian.

Notes:

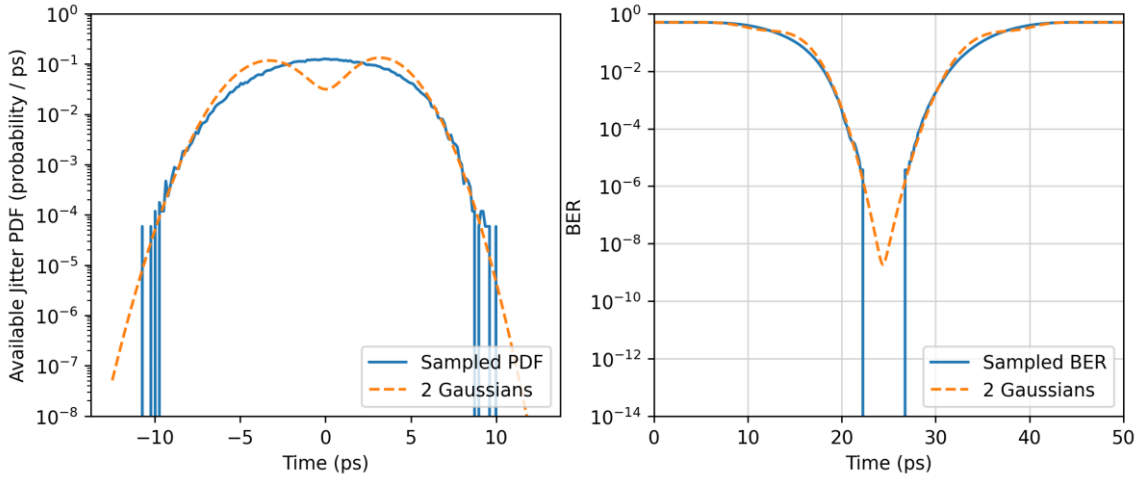
- To simulate jitter, the length of bit pattern should be 2^{20} or longer. This will give you some PDF data at regions where the probability is $2^{-20} \approx 10^{-6}$.
- For jitter analysis purpose, you should not add 10^{-15} to the array during constructing statistical eye, because it will affect the jitter distribution at low probabilities. Simply construct the eye as *it is*. You may add 10^{-15} or any small number to it when plotting, to avoid taking logarithm of 0.
- During fitting, you may look at the PDF and BER both, because sometimes it is unclear whether the fitting is good just from one plot. Notice that the *left* part of PDF corresponds to the *right* part of BER, and vise versa.

Optional problem: Simulate the circuit in ADS ChannelSim to verify your results.

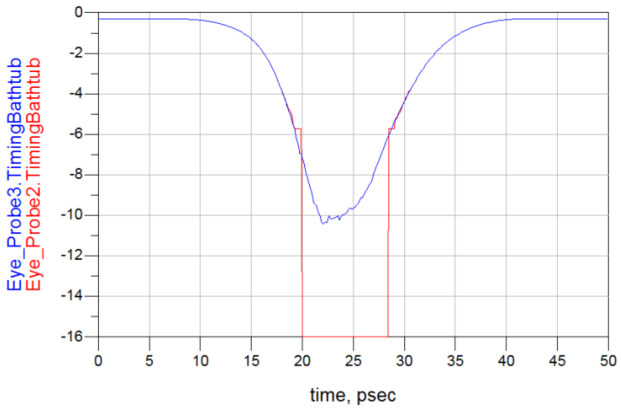
Appendix 1: Example answer

<table>
<tr><th>CDF</th><th>BER</th></tr>
<tr><td> 
</td><td> 
</td></tr>
<tr><td colspan="2">Case 3 Rx Python</td></tr>
<tr><td colspan="2"> 
</td></tr>
</table>

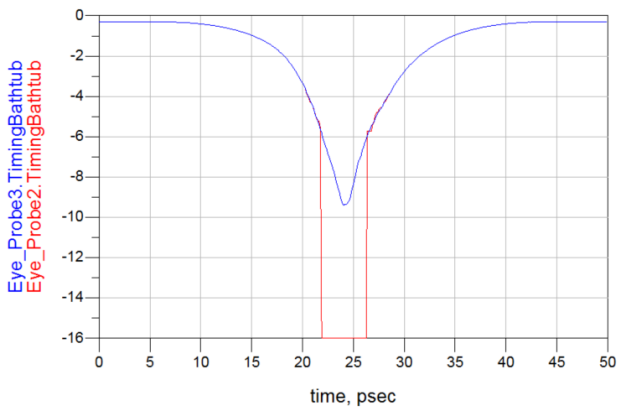
Case 4 Rx Python



Case 3 ADS



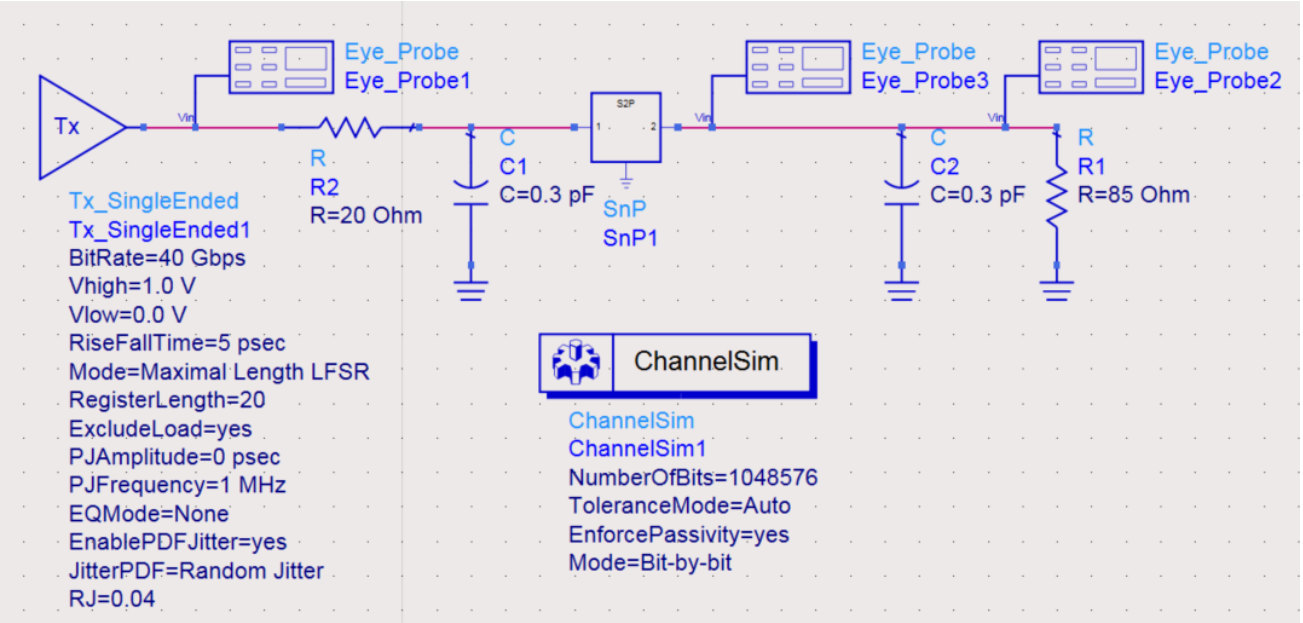
Case 4 ADS



Note: For ADS's Case 3 results, the behavior at BER = 10^{-10} is strange, isn't it? Is it reliable?

Appendix 2: ADS setting

To simulate the eyes in ADS channelsim, connect the circuit as below.



Note that the unit of RJ is UI, so σ = 1 ps = 0.04 UI for 40 Gbps signal.

In the Eye_Probe setting, select or unselct the “Extrapolate in bit-by-bit mode”, and add the measurement of “Bathtub”.

